On the Exploitability of Fair Multi-Agent Reinforcement Learning

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***Abstract*—Fair cooperative multi-agent reinforcement learning (MARL) methods such as SOTO and FEN train agents to optimize a fairness-inducing welfare function (e.g. the Generalized Gini Welfare) over the team's returns, so that no agent is left behind. These methods implicitly assume that *every* agent cooperates in being fair. Building on the known observation that welfare-*maximizing* cooperation can be gamed by defectors, we isolate and quantify the *fairness*-specific case: a single self-interested agent can exploit a welfare-optimizing team, because fair agents sacrifice their own utility to raise the worst-off and a best-response defector appropriates that surplus. In rivalrous allocation games the effect is *exact by construction*: when the welfare-optimal policy is to yield, a lone claimant takes everything, so the free-ride factor equals the team size N (these serve as analytical sanity checks, and a faithful SOTO learner reproduces the same $N\times$ outcome, confirming it learns the yield policy). Our empirical evidence comes from settings with genuine learned contention: a collision-prone claim game ($3.0\times$) and a FEN-style Matthew-effect gridworld ($1.9\times$, with seed variance), where the Jain index falls sharply. We further show that a natural *policy-level* defense (reciprocity with adversarial co-training) *fails*—provably, in the all-or-nothing model—because cooperators cannot deny a free-rider without wasteful levelling-down. Finally we show that an egalitarian (worst-off-first) allocation rule, EKREM, restores incentive-compatibility (free-ride ≈ 1 for $N \geq 4$) by taking allocation *out of the agents' hands*; we are explicit that this sidesteps, rather than solves, the harder problem of *decentralized* robustness.**

***Keywords*—multi-agent reinforcement learning, fairness, exploitability, incentive compatibility, social welfare**

I. Introduction

Cooperative multi-agent reinforcement learning (MARL) increasingly targets not only efficiency but *fairness*: in traffic control, resource allocation, and networking, optimizing aggregate return alone can starve individual agents. Leading fair-MARL methods—FEN [1] and SOTO [2]—reframe the objective as a fairness-inducing welfare function over the agents' returns, typically the Generalized Gini Welfare (GGF), which weights worse-off agents more heavily and thus rewards equity as well as efficiency.

These methods are designed and evaluated under an implicit assumption: that all N agents are jointly optimizing the shared welfare. In many deployments this is optimistic. Consider fair traffic-signal control where one intersection's controller is upgraded to greedily minimize its own delay, or fair bandwidth sharing where one client maximizes its own throughput: the remaining controllers still "play fair," yielding capacity to the worst-off. Agents may be controlled by different stakeholders, or one agent may simply be trained (or retrained) to maximize its own return. We ask a basic robustness question: *what happens to a welfare-optimizing fair team when a single member is self-interested?* That welfare-*maximizing* cooperation can in general be gamed by defectors was noted by Ivanov et al. [3]; our focus is the *fairness* instance specifically—its mechanism, its quantification, and what it takes to fix.

We find the answer is sharp. Because fair (team-oriented) agents *sacrifice* their own utility—yielding resources to raise the worst-off—a self-interested agent can free-ride on that sacrifice. We make this concrete with a best-response analysis and report three contributions:

- **A diagnostic.** A single best-response defector exploits a GGF-welfare team, collapsing the Jain index toward $1/N$. In the yield-based allocation games the over-share is $\rho{=}N$ *by arithmetic* (analytical sanity checks; a faithful SOTO learner reproduces the same $N\times$, confirming it learns the welfare-optimal yield policy). The substantive evidence is in settings with *learned* contention—a collision-prone claim game ($3.0\times$) and a Matthew-effect gridworld ($1.9\times$, real seed variance).
- **A negative result.** The obvious policy-level defense—reciprocity with adversarial co-training—fails. We prove (for the all-or-nothing model) that cooperators cannot deny a free-rider efficiently; denial requires wasteful levelling-down, which an efficiency–equity objective rightly refuses.
- **A positive result (with a caveat). EKREM** instantiates the classic egalitarian (worst-off-first) allocation rule inside the MARL loop: resources go by *need*, so greedy claims earn nothing and incentive-compatibility is restored (free-ride ≈ 1 for $N \geq 4$). We are explicit that it works by *removing agents' control over allocation*—it sidesteps, not solves, decentralized robustness.

II. Background and Related Work

Welfare-based fair MARL. Given per-agent (discounted) returns $\mathbf{J} = (J_1, \ldots, J_N)$, fair-MARL maximizes a Schur-concave welfare $W(\mathbf{J})$ that is increasing (efficiency) and favors equity. A common choice is the Generalized Gini Welfare $W(\mathbf{J}) = \sum_k w_k J_k^{\uparrow}$, where $J^{\uparrow}$ sorts returns ascending and w_k is decreasing, so the worst-off receive the largest weight [4]. FEN [1] uses a hierarchical controller over sub-policies driven by a fair-efficient reward; SOTO [2] learns a *Self-Oriented* sub-network (own return) and a *Team-Oriented* sub-network (welfare), annealing behavior from self-oriented to team-oriented over training.

Self-interest and mediators. The closest work is the mediator line. Ivanov et al. [3] explicitly observe that welfare-*maximizing* cooperation can be exploited by selfish defectors and restore cooperation through an incentive-compatible (IC) mediator that respects agents' boundaries; Dodwadmath and Maghsudi [5] elicit fair leaders via mediators (their JAM-QL framework). *We credit [3] for the general observation* that welfare-maximizing teams are exploitable—it is not our contribution. What is ours is specific to it: (i) the *fairness/GGF instantiation*, where the equity term that raises the worst-off is precisely the appropriable surplus; (ii) a *quantified* best-response exploitability—the free-ride factor ρ measured against a faithful fair learner (SOTO) and its scaling $\rho \approx N$; and (iii) a *policy-vs-mechanism dichotomy*: we prove (Prop. 1) that learned reciprocity cannot fix it, so the remedy must be a need-based allocation mechanism (EKREM), which complements rather than reuses [3]'s mediator.

Social dilemmas and free-riding. The vulnerability we expose is a fairness-specific instance of free-riding in sequential social dilemmas [6] and common-pool problems: a contributor's sacrifice can be appropriated by a non-contributor. Whereas social-dilemma MARL studies the *emergence* of cooperation, we study how an explicitly fairness-seeking objective *creates* an appropriable surplus—the utility team-oriented agents forgo to raise the worst-off—and who captures it.

Mechanism design and fair division. Our fix is not a new allocation rule but the import of a classic one. Allocating to the worst-off is the egalitarian / leximin principle of fair division [7], and making truthful (here, need-revealing) behavior optimal is the goal of mechanism design [8]. EKREM instantiates this known incentive-compatible egalitarian mechanism *inside* the MARL loop.

Exploitability and best response. In competitive MARL, a policy's *exploitability* is the gain of a best response to it. We import this lens to *cooperative* fairness: we hold the fair team fixed and train a best-response defector, treating its realized over-share as an exploitability measure for the fairness mechanism.

III. Problem Formulation

We consider N agents acting over an episode of T steps. Let $u_i \geq 0$ be agent i's accumulated utility (resources collected) and $\mathbf{u} = (u_1, \ldots, u_N)$. A *fair* team maximizes a Schur-concave welfare $W(\mathbf{u})$ that is increasing in each u_i (efficiency) and favors equity. We use the Generalized Gini Welfare

$$W(\mathbf{u}) = \sum_{k=1}^{N} w_k \, u_k^{\uparrow}, \qquad w_1 > w_2 > \cdots > w_N > 0, \quad (1)$$

where $u^{\uparrow}$ sorts $\mathbf{u}$ ascending, so the worst-off agent carries the largest weight. We report fairness with Jain's index [9], $\mathrm{Jain}(\mathbf{u}) = \left(\sum_i u_i\right)^2 / \left(N \sum_i u_i^2\right) \in (0, 1]$ (1 = perfectly fair), and the Gini coefficient.

Threat model. A fair team of N policies is trained to maximize $W(\mathbf{u})$. We then *freeze* the team, replace one agent by a *defector*, and train the defector to maximize its *own* utility u_{def} as a best response to the frozen teammates. We quantify the exploit by the *free-ride factor*

$$\rho = \frac{u_{\text{def}}}{\bar{u}}, \qquad \bar{u} = \tfrac{1}{N} \textstyle\sum_i u_i, \quad (2)$$

the defector's utility over the equal (fair) share. $\rho = 1$ is exactly fair; $\rho = N$ means the defector took everything. A fairness method is *exploitable* if a best-response defector attains $\rho \gg 1$, and *robust* (incentive-compatible) if $\rho \approx 1$.

IV. Environments

Allocation games and a Matthew-effect gridworld. *(i) claim*: each step one resource is available; agents choose CLAIM/YIELD; a sole claimer wins it, collisions waste it. *(ii) donate*: as claim, but if all yield the resource auto-routes to the worst-off (so fair play is to yield and let the needy collect). *(iii) matthew*: among claimers the *richest* wins, capturing FEN's rich-get-richer dynamic. *(iv) FEN gridworld*: a 2D grid with one (respawning) resource; an agent's movement speed scales with its wealth (rich move every step, poor intermittently), so fair play requires the rich to yield.

In every game, action 1 is the *self-interested* act (claim/keep) and 0 is to *defer* (yield/donate). Under GGF the optimal joint policy is the same across games—"act selfishly only when you are the worst-off"—which a fair team must learn, and which a defector violates.

V. Experimental Setup

Fig. 1 gives an overview of the policy architecture and the allocation mechanism. **Fair team.** We train two fair learners. (i) A simplified shared-policy GGF learner: a single policy controls all agents and is optimized to maximize $W(\mathbf{u})$. (ii) A faithful **SOTO**: a Self-Oriented sub-network (own-return objective) and a Team-Oriented sub-network (GGF objective), with the behavior probability annealing from self-oriented to team-oriented over training; at convergence (β=0) the team is purely team-oriented, i.e. fair.

Defector. After the fair team converges we freeze it, designate one agent as the defector, and train its policy to maximize its own utility against the frozen teammates—a best response.

Optimization. All policies are two-layer MLPs trained with REINFORCE [10] (Adam, 512 parallel episodes). Each agent observes its own and the team's utility statistics (its rank/being-worst-off) and, in the gridworld, its position relative to the resource. The gridworld uses a dense GGF-increment reward ($r_t = W(\mathbf{u}_t) - W(\mathbf{u}_{t-1})$) so that navigation is learnable; the abstract games use the terminal welfare. Unless noted, N=4 and results are averaged over 3 seeds.

VI. Welfare-Optimizing Fairness is Exploitable

Table I reports the exploit at N=4 (Fig. 2 visualizes it against EKREM). We separate two kinds of result.

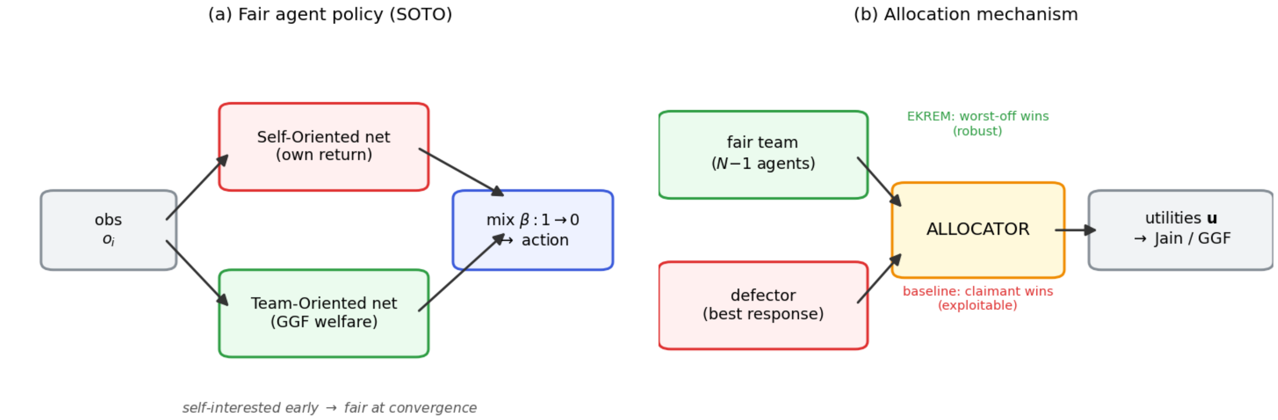

Fig. 1: Overview. (a) The fair agent policy: a SOTO learner combines a Self-Oriented sub-network (own return) and a Team-Oriented sub-network (GGF welfare), annealing from self-interested to fair over training. (b) The allocation mechanism turns agents' requests into utilities: the *baseline* awards the resource to the claimant (exploitable by a defector), whereas **EKREM** awards it to the worst-off (need-based, robust).

TABLE I: Exploitability at $N{=}4$. A single best-response defector vs. a trained fair team. Free-ride = defector utility / fair share.

Environment	Fair-team Jain	Free-ride	Jain w/ defector
donate	1.00	4.0×	0.25
claim	0.998	3.0×	0.43
matthew	1.00	4.0×	0.25
FEN gridworld	0.88	1.9×	0.67
faithful SOTO (donate)	1.00	4.0×	0.25

Analytical sanity checks (donate, matthew, SOTO). In the yield-based games the welfare-optimal joint policy is for everyone to yield (the resource auto-routes to the worst-off). A defector that always claims is then the *sole* claimant and takes the resource every step, so $\rho{=}N$ *by arithmetic*—not by learning. This is exactly why these entries are zero-variance. The faithful *SOTO* run is the same fact reproduced by a real learner: its 4.0× confirms that SOTO converges to the GGF-optimal yield policy, *not* that the exploit is independently nontrivial. We therefore treat these as sanity checks rather than empirical evidence.

Empirical evidence (claim, gridworld). The substantive results come from settings with genuine learned contention. In *claim* (no auto-routing, collisions waste the resource) the defector reaches 3.0× rather than 4.0×: on steps where the worst-off cooperator also claims, the collision wastes the resource and the defector *forfeits* it, so the number is shaped by the contention dynamics, not just N. The *FEN gridworld* (1.9×, with real seed variance) is the strongest test: agents must navigate, the defector cannot be everywhere, and fairness still falls from 0.88 to 0.67.

The exploit scales with team size. Fig. 3 shows that in the donate game the free-ride factor grows essentially as N (the defector takes everything, so its share is $N\times$ fair) and the post-defector Jain falls as $1/N$: at $N{=}16$ a single self-interested agent drives fairness to 0.06. The larger and more cooperative the team, the more there is to exploit.

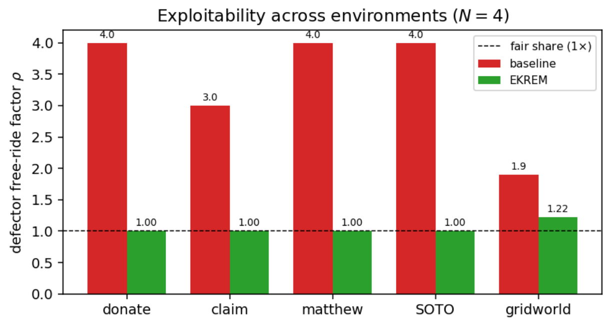

Fig. 2: Defector free-ride factor ρ per environment ($N{=}4$). A single self-interested agent free-rides 1.9–4× against the welfare-fair baseline; EKREM brings every case back toward the fair share ($\rho \approx 1$).

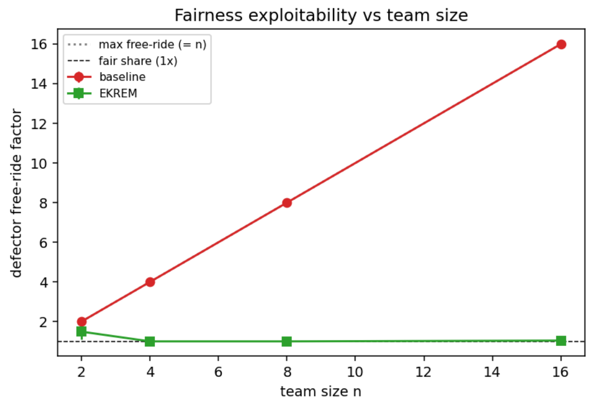

Fig. 3: Free-ride factor vs. team size N. The baseline welfare-fair team is exploited along the diagonal (free-ride $\approx N$); EKREM (our need-based allocator) stays flat at ≈ 1. Mean over 3 seeds.

VII. Why a Policy-Level Defense Fails

A natural fix is to make cooperation *conditional*: agents detect an over-claimer (a reciprocity feature) and are adversarially co-trained against a best-response defector so they learn to contest it. This *fails*: a freshly trained defector still free-rides $\approx 4\times$ and the contesting degrades the team's own fairness.

The reason is structural, and we can state it precisely.

TABLE II: Scaling with team size N (donate game, 3 seeds). The baseline exploit grows as $\approx N$ and fairness collapses as $\approx 1/N$; EKREM holds $\rho \approx 1$ and Jain ≈ 1 for $N \geq 4$, with a residual at N=2 (ρ=1.49; see text).

	Baseline		EKREM	
N	free-ride ρ	Jain	free-ride ρ	Jain
2	2.00	0.50	1.49	0.77
4	4.00	0.25	1.00	1.00
8	8.00	0.125	1.00	1.00
16	16.00	0.063	1.04	0.998

EKREM allocation step
Input: utilities $\mathbf{u}$; requesters C; fair share $s = t/N$
1. $E \leftarrow \{i \in C : u_i < s\}$ (requesting *and* below fair share)
2. **if** $E \neq \emptyset$ **then** $w \leftarrow \arg\min_{i \in E} u_i$ (neediest eligible)
3. **else** $w \leftarrow \arg\min_i u_i$ (worst-off overall; no waste)
4. $u_w \leftarrow u_w + 1$

Fig. 4: The EKREM allocation step: resources flow by *need*, so a claim succeeds only when the claimer is genuinely worst-off.

Proposition 1 (Futility of policy-level defense): Consider a rivalrous resource allocated each step to a claimant (sole claimant wins; a collision wastes the resource), against a defector that claims at every step. Then for *any* cooperator policies, at each step either (i) some cooperator claims, colliding with the defector so the resource is wasted, or (ii) no cooperator claims and the defector wins. In both cases the cooperators obtain zero of that resource. Hence the only way to reduce the defector's utility is to force collisions, which wastes the resource without benefiting cooperators; a GGF-maximizing team is therefore indifferent between yielding and contesting.

Intuitively, cooperators cannot *capture* the surplus they would deny the defector—denial is pure levelling-down, which an efficiency–equity objective correctly refuses. Policy-level reciprocity thus has no efficient leverage; robustness must come from the allocation *mechanism*, which can redirect the contested resource to a needy cooperator rather than wasting it. We emphasize that the proposition is specific to *all-or-nothing waste*; under partial allocation or graded contention the "collision $\Rightarrow$ total waste" premise breaks and the conclusion need not hold (Sec. IX).

VIII. EKREM: A Need-Based Fair Allocator

EKREM instantiates the classic egalitarian (worst-off-first / leximin) allocation rule [7] inside the MARL loop. Rather than "whoever grabs the resource gets it," the resource is allocated by *need*: to the worst-off agent (overall, or among claimers). The rule itself is not new—it is the standard incentive-compatible egalitarian mechanism [8]—and that is precisely the point. A defector's greedy claims are honored only when it is genuinely worst-off (exactly when receiving is fair), so over-claiming earns nothing, while efficiency is preserved because the resource still goes to a cooperator. Fig. 4 states the rule. We stress that EKREM achieves this by *removing agents' control over allocation*: once a central need-based allocator decides, policies no longer determine who receives the resource, so there is nothing left to exploit (see Sec. IX).

Table III shows EKREM holds the free-ride factor at ≈ 1 and Jain at ≈ 1 across the abstract environments and against faithful SOTO, and substantially mitigates the gridworld exploit (1.90$\rightarrow$1.22, fair-team Jain preserved at 0.85, three seeds; per-seed in Table IV). Because allocation tracks *need* rather than *grabbing*, a defector's claims are honored only when it is genuinely worst-off—exactly when receiving is fair—so over-claiming earns nothing and incentive-compatibility is tight for $N \geq 4$ (ρ≈1, Table II). The exception is N=2 (ρ=1.49, Jain 0.77): with only two agents a defector can deliberately drive itself below the fair share, become the genuine worst-off, claim and overshoot before the partner recovers, so the need-gate oscillates rather than binding tightly. We report N=2 as a residual (as with the gridworld); the mechanism is tight for $N \geq 4$.

TABLE III: EKREM restores incentive-compatibility. Free-ride factor with a best-response defector (lower is better; 1 = fair).

Environment	Baseline	EKREM
donate / matthew (abstract)	4.0×	1.0×
faithful SOTO	4.0×	1.0×
FEN gridworld (3 seeds)	1.90×	1.22×

IX. Discussion and Limitations

Centralization. EKREM allocates by need and therefore requires (an estimate of) agents' utilities—the same global-information assumption already made by SOTO's welfare gradient. It effectively moves fairness from the *policy* to the *mechanism*: rather than hoping self-interested agents behave fairly, the allocation rule makes greedy behavior unrewarding. Where a trusted allocator (a scheduler, a base station, a market clearing house) exists, this is a practical and provably incentive-compatible fix.

The hard open problem. Proposition 1 applies to the all-or-nothing rivalrous model: a contested resource is either won by a single agent or wasted. Under that model a decentralized cooperator cannot deny a free-rider except by wasteful levelling-down, which an efficiency–equity objective refuses. We *conjecture*, but do not prove, that the difficulty extends more broadly; with partial allocation or graded contention resolution the premise breaks and decentralized cooperators may regain leverage (e.g., by partially contesting). Making welfare-fair teams non-exploitable with only local information and action is, to our knowledge, open; we offer the centralized EKREM as an upper bound that decentralized methods should aim to match.

Scope. Our environments are deliberately controlled—small resource-allocation games and a compact gridworld—to isolate the mechanism; larger gridworlds, continuous control, and multiple simultaneous defectors are natural extensions, as is a coalition of colluding defectors. We expect the qualitative story to persist: any objective that has cooperators sacrifice for the worst-off creates an appropriable surplus.

Takeaway for fair-MARL evaluation. Fair-MARL methods are typically evaluated on the fairness they *achieve under full cooperation*. We argue they should additionally be reported on the fairness they *retain under self-interest*—an exploitability audit—since the former can be high while the latter is near-zero.

X. Conclusion

Welfare-optimizing fair MARL is exploitable: a single best-response defector free-rides by up to $N\times$—exact in the yield-based games, and 1.9–$3.0\times$ under genuine learned contention—the effect worsens with team size, and it holds against a faithful SOTO learner. That welfare-maximizing cooperation is gameable in general was observed by Ivanov et al. [3]; our contribution is the fairness-specific instance, its quantified best-response free-ride factor, and a policy-vs-mechanism dichotomy—learned reciprocity provably cannot fix it (Prop. 1), whereas EKREM can, by relocating allocation to a trusted need-based mechanism, *sidestepping rather than solving* the decentralized case. We hope the exploitability lens encourages fair-MARL methods to be evaluated not only for the fairness they achieve under full cooperation, but for the fairness they *retain* under self-interest.

Code Availability

All code (environments, SOTO, EKREM, and experiments) and the LaTeX source are available at https://github.com/highcansavci/ekrem-fair-marl.

Appendix A
Implementation Details

All policies are two-layer $\tanh$ MLPs (width 32 for the abstract games, 64 for the gridworld) trained with REINFORCE and Adam (learning rate 3×10^{-3}) using B=512 parallel episodes. Episodes last T=100–200 (abstract games) or T=80 (gridworld). The GGF weights are $w_k \propto (N-k+1)$ over ascending-sorted utilities, placing the largest weight on the worst-off. The fair team trains for 400–800 updates and the best-response defector for the same. The faithful SOTO uses separate Self- and Team-Oriented MLPs and anneals the self/team mixing probability β from 1 to 0 over the first 70% of training; we evaluate at β=0 (purely team-oriented). All reported numbers average 3 seeds.

Appendix B
Per-seed Results

The abstract games and the faithful SOTO are effectively deterministic at convergence (zero seed variance), which is why Tables I and III report exact $4.0\times$/$1.0\times$ values. The gridworld, with stochastic movement and respawns, varies mildly; Table IV gives the per-seed free-ride factor, confirming the exploit ($\approx 1.9\times$) and EKREM's mitigation ($\approx 1.2\times$) are reproducible.

TABLE IV: FEN gridworld free-ride factor ρ per seed (N=4).

Allocator	seed 0	seed 1	seed 2	mean
Matthew (baseline)	1.91	1.91	1.87	1.90
EKREM (need-based)	1.17	1.25	1.24	1.22

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